Performance Testing

Date	30 September 2026
Team ID	SWTID-2026-9166
Project Name	AI FitTrack
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	JMeter
Type of Testing	Load Testing, Stress Testing, Spike Testing
Target Module / API	AI BlogNest API – Blog Creation, Retrieval and AI Content Generation APIs
Test Environment	Local / Staging Server
Test Date	30 September 30, 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Normal Load – Users accessing blog APIs	50	60	API should respond within 2 seconds
2	High Load – Multiple users creating and retrieving blogs	200	120	API should remain stable with acceptable response time
3	Stress Test – Maximum concurrent API requests	500	180	System should remain available with only minor delays
4	Spike Test – Sudden increase in API requests	1000	60	API should handle sudden traffic without crashing

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.4 seconds	Pass	API responded quickly under normal load
2	Response Time (Max)	< 5 seconds	4.2 seconds	pass	Slight increase observed under heavy load
3	Throughput (Req/sec)	> 100 req/sec	145 req/sec	pass	Good request handling capacity
4	Error Rate	< 1%	0.6%	pass	Errors remained within acceptable limit
5	CPU Utilization	< 80%	68%	pass	CPU usage remained stable
6	Memory Utilization	< 80%	71%	pass	Memory consumption was within limit

Step 4: Observations & Analysis

Key Findings:

- AI BlogNest API performed well under normal and high user loads.
- Average API response time remained below the target of 2 seconds.
- Throughput remained stable as the number of concurrent users increased.
- Error rate remained below 1%.
- CPU and memory utilization stayed within acceptable limits.
- The system remained stable during stress and spike testing.

Bottlenecks Identified:

- Response time increased slightly during peak concurrent requests.
- AI content generation requests consumed more processing time compared with normal blog retrieval requests.
- Database/API processing may become a bottleneck under very high traffic.

Optimization Steps Taken:

- Optimized API request and response handling.
- Reduced unnecessary database queries.
- Added caching for frequently accessed blog data.
- Improved database query performance.
- Recommended API rate limiting and connection pooling for higher loads.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):



